

$$f : \mathbb{R}^2 \rightarrow \mathbb{R}$$

$$\begin{aligned} \max f(x) &= 30x_1 + 20x_2 \\ 2x_1 + x_2 &\leq 40 \\ 3x_1 + x_2 &\leq 24 \\ x_1, x_2 &\geq 0 \end{aligned}$$

Calcolo dei costi

$$c_i = c_N + c_B^T \cdot B_N^{-1}$$

$$c_1 = -30 - (0;0) \cdot \begin{pmatrix} 2 \\ 3 \end{pmatrix} = -30$$

$$c_2 = -20 - (0;0) \cdot \begin{pmatrix} 1 \\ 1 \end{pmatrix} = -20$$

$$c_3 = 0 - (0;0) \cdot \begin{pmatrix} 1 \\ 0 \end{pmatrix} = 0$$

$$c_4 = 0 - (0;0) \cdot \begin{pmatrix} 0 \\ 1 \end{pmatrix} = 0$$

$$c_5 = 0 - (0;0) \cdot \begin{pmatrix} 40 \\ 24 \end{pmatrix} = 0$$

Metodo del simpleso

	x_1	x_2	s_1	s_2	b
s_1	2	1	1	0	40
s_2	3	1	0	1	24
	-30	-20	0	0	0

Entra x_1 ed esce s_2 :

	x_1	x_2	s_1	s_2	b
s_1	2	1	1	0	40
s_2	3	1	0	1	24
	-30	-20	0	0	0

$$R3 \leftarrow R3 + R2 \cdot 10$$

$$\begin{array}{cccc|c} -30 & -20 & 0 & 0 & 0 \\ 30 & 10 & 0 & 10 & 240 \\ \hline 0 & -10 & 0 & 10 & 240 \end{array}$$

↓

	x_1	x_2	s_1	s_2	b
s_1	2	1	1	0	40
s_2	3	1	0	1	24
	0	-10	0	10	240

Forma Standard

$$\begin{aligned} \min f(x) &= -30x_1 - 20x_2 \\ 2x_1 + x_2 + s_1 &= 40 \\ 3x_1 + x_2 + s_2 &= 24 \\ x_1, x_2, s_1, s_2 &\geq 0 \end{aligned}$$

$$R2 \leftarrow R2 \cdot \frac{1}{3}$$

	x_1	x_2	s_1	s_2	b
s_1	2	1	1	0	40
s_2	1	$\frac{1}{3}$	0	$\frac{1}{3}$	8
	0	-10	0	10	240

$$R1 \leftarrow R1 - R2 \cdot 2$$

$$\begin{array}{cccc|c} 2 & 1 & 1 & 0 & 40 \\ 2 & \frac{2}{3} & 0 & \frac{2}{3} & 16 \\ \hline 0 & -\frac{1}{3} & 1 & -\frac{2}{3} & 24 \end{array}$$

↓

	x_1	x_2	s_1	s_2	b
s_1	0	$-\frac{1}{3}$	1	$-\frac{2}{3}$	24
x_1	1	$\frac{1}{3}$	0	$\frac{1}{3}$	8
	0	-10	0	10	240

Entra x_2 ed esce x_1 :

	x_1	x_2	s_1	s_2	b
s_1	0	$-\frac{1}{3}$	1	$-\frac{2}{3}$	24
x_1	1	$\frac{1}{3}$	0	$\frac{1}{3}$	8
	0	-10	0	10	240

$$R1 \leftarrow R1 + R2$$

$$\begin{array}{cccc|c} 0 & -\frac{1}{3} & 1 & -\frac{2}{3} & 24 \\ 1 & \frac{1}{3} & 0 & \frac{1}{3} & 8 \\ \hline 1 & 0 & 1 & -\frac{1}{3} & 16 \end{array}$$

↓

	x_1	x_2	s_1	s_2	b
s_1	1	0	1	$-\frac{1}{3}$	16
x_1	1	$\frac{1}{3}$	0	$\frac{1}{3}$	8
	0	-10	0	10	240

$$R2 \leftarrow R2 \cdot 3$$

	x_1	x_2	s_1	s_2	b
s_1	1	0	1	$-\frac{1}{3}$	16
x_1	3	1	0	1	24
	0	-10	0	10	240

$$R3 \leftarrow R3 + R2 \cdot 10$$

$$\begin{array}{cccc|c} 0 & -10 & 0 & 10 & 240 \\ 30 & 10 & 0 & 10 & 240 \\ \hline 30 & 0 & 0 & 20 & 480 \end{array}$$

↓

	x_1	x_2	s_1	s_2	b
s_1	1	0	1	$-\frac{1}{3}$	16
x_2	3	1	0	1	24
	30	0	0	20	480

$$x_1 = 0$$

$$x_2 = 24$$

$$s_1 = 16$$

$$s_2 = 0$$

$$\min f(x) = -30x_1 - 20x_2 \rightarrow f(x) = -30(0) - 20(24) = 0 - 480 = -480$$

$$\max f(x) = 30x_1 + 20x_2 \rightarrow \boxed{480}$$

Metodo Grafico



Figura 1: La retta $x_2 = -\frac{3}{2}x_1 + 24$ passa per il punto di ottimo.